

Antoine EDY

AI Research Scientist

antoineedy@outlook.fr

Paris, France

Education

- **University of Surrey** — *Master of Science in Artificial Intelligence* *2023-2024*
  - International Excellence Scholarship recipient. Graduated with Highest Honors
  - NLP, CV, Audio Processing, Symbolic Knowledge Representation and Logical Reasoning, People-Centered AI
  - Master's Thesis: *Zero-shot and Few-shot Adaptation of Multimodal Foundation Models*
- **École Centrale de Lyon** — *General Engineering Program (Grande École)* *2021-2024*
  - Specializations: Mathematics, Machine Learning, Probability Theory, Chaos and Fractals
  - Project Leader: Unsupervised MCU Classification with Deep Learning for STMicroelectronics
- **Collège Stanislas Paris VI** — *Classe Préparatoire, MPSI-MP* *2019-2021*

Experience

- **Illuin Technology** — *AI Research Scientist* *September 2024 - Present*
  - Co-led the creation of ViDoRe V3, establishing the gold standard benchmark for multimodal document retrieval evaluation; contributor to the MTEB ecosystem
  - Designed an agentic harness that writes and retrieves memory for continuous in-context learning on long-horizon tasks
  - Conducted research on Multimodal Retrieval and late interaction dynamics (ColBERT/ColPali-like models)
  - Fine-tuned LLMs for tool calling in agentic RAG contexts; fine-tuned speech-to-text models (Whisper, NVIDIA Parakeet) for domain-specific industrial applications
  - Conducted 50+ technical interviews as part of the hiring process for Data Scientist positions
- **Centre for Vision, Speech and Signal Processing (University of Surrey)** — *Research Intern* *2024*
  - Studied zero-shot semantic segmentation using multimodal foundation models; improved ZegCLIP baseline with multi-scaling techniques and inner encoder layers
  - Research led to Master's thesis: *Exploring Zero-Shot Capabilities of Multimodal Foundation Models for Semantic Segmentation*, supervised by Prof. Muhammad Awais Rana and Dr. Srinivasa Rao Nandam 🔗 ResearchGate
- **AI-vidence** — *ML Research Intern* *2023*
  - Developed a Python library for tabular model explainability and cluster analysis
  - Presented at SpringTech Paris-Saclay 2023 at HEC Paris

Publications

⋆ *denotes equal contribution*

- **ViDoRe V3: A Comprehensive Evaluation of RAG in Complex Real-World Scenarios**
  - A. Loison⋆, Q. Macé⋆, **A. Edy**⋆, V. Xing, T. Balough, G. Moreira, B. Liu, M. Faysse et al.
  - Multimodal RAG benchmark: 26,000 pages, 18,594 queries across 10 domains in 6 languages, 12,000h of human annotation. Work made in collaboration with NVIDIA.
  - **ACL 2026 Main Conference** (Oral Presentation), 🔗 arXiv:2601.08620
- **Self-Supervised Heuristic Extraction and Retrieval for Planning Agents**
  - **A. Edy**⋆, M. Conti⋆, V. Xing, N. Benhamdane, M-A. Allard, G. Viaud
  - Memory generation for in-context learning without any training data: the agent invents its own tasks and generates actionable heuristics. SOTA among training-free memory methods on AppWorld and τ^2-bench.
  - **In Submission**
- **Analyzing Targeted Behaviors of Late Interaction Models**
  - **A. Edy**⋆, M. Conti⋆, Q. Macé⋆
  - Analysis of length bias in multi-vector scoring and MaxSim similarity distribution on NanoBEIR benchmark
  - **ECIR 2026** Late Interaction Workshop (Poster), 🔗 arXiv:2603.26259

Talks & Blog Posts

- **Multimodal Retrieval and ViDoRe V3**, Invited talk, Neurons & Peppers (Datacraft, Paris) *2026*
- **Agentic Search Framework for ViDoRe V3** 🔗 Hugging Face blogpost *2026*
- **Xplique: analysing different scales of interpretability** 🔗 AI-vidence blogpost *2023*

Technical Skills

ML Frameworks: PyTorch, Hugging Face (Transformers, PEFT, Accelerate), W&B, NVIDIA NeMo, OpenMMLab
Information Retrieval: SentenceTransformer, pylate, colpali-engine, MTEB, FAISS, (Fast)PLAID, ElasticSearch
Infra. and Ops.: Kubernetes, Docker, vLLM, SLURM, HTCondor, skypilot, GCP
Languages: French (Native), English (C1, TOEFL 623/677), Spanish (B1)

Hobbies

- Competitive tennis, football, and any sport going (any day, any time)
- Chess, where my tactics and my blunders are evenly matched
- Cinema, literature, hip-hop music
- Digital graphic design and art culture